Describe Data by Mean Absolute Deviation

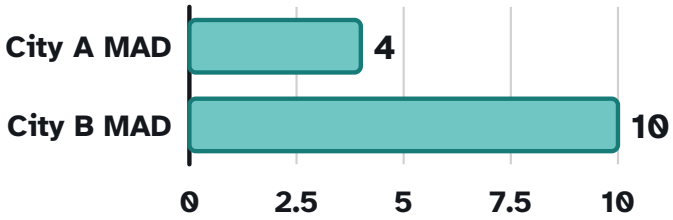
Use the MAD to compare two data sets that have the same mean.

Name _____ **Date** _____

Tonight's spark

Two cities both have an average high of 52 degrees. City A highs: 46, 50, 54, 58. City B highs: 36, 48, 56, 68. You pack one jacket. Which city is easier to pack for?

Guess first: which city has highs that stay closer to 52?



Our class model: comparing with the MAD

Try it

Work it out on paper. Write one answer on each line.

1 **A large MAD means the values are...**

A. Scattered far from the mean **B.** Close to the mean **C.** All equal
D. Greater than the mean

My answer _____

2 **Set A is 9, 10, 11. Set B is 1, 10, 19. Both have mean 10. Which statement is true?**

A. The sets are the same because the means match
B. Set B is more spread out than Set A, even with the same mean
C. Set A is more spread out than Set B **D.** Neither set has any spread

My answer _____

3 **Set M is 7, 9, 11, 13 with mean 10. What is its MAD?**

A. 8 **B.** 10 **C.** 2 **D.** 3

My answer _____

4 **Set N is 4, 8, 12, 16 with mean 10. What is its MAD?**

A. 4 **B.** 16 **C.** 2 **D.** 12

My answer _____

5 **Set M has MAD 2 and Set N has MAD 4, and both have mean 10. Which statement is true?**

A. Set M varies more because its MAD is smaller

B. Set N varies more because its MAD is larger

C. They vary equally because the means are equal **D.** The sets cannot be compared

My answer _____

Talk about it

The means match. How does the MAD show the difference?

Home mission

Two bus routes both average 20 minutes. Route 1 takes 19, 20, 20, 21. Route 2 takes 10, 15, 25, 30. Ask your student which route they want on a school morning, and why.

If we get stuck, ask

- How far is this value from the mean? Distances cannot be negative. A value equal to the mean has distance 0.
- Did you average the distances, or the original values?
- Same mean — does that mean the two sets look the same?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/2-9-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it